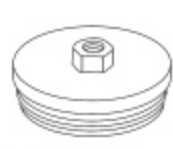


Crankshaft Rear Seal with Retainer Plate



Special Tool(s)

 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST2980-A	Installer, Rear Main Seal 303-1250
 ST1382-A	Remover, Crankshaft Rear Oil Seal 303-519 (T95P-6701-EH)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)

Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS- M2C945-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4

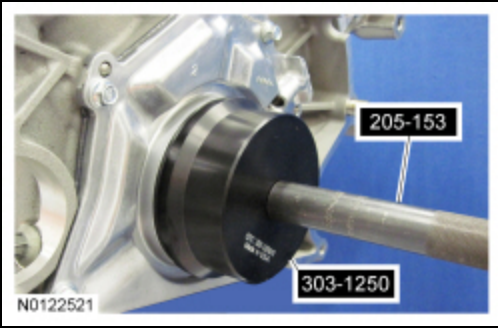
Removal

1. Remove the flexplate. For additional information, refer to Flexplate or Flywheel and Crankshaft Rear Seal — Exploded View and Flexplate.
2. Remove the engine separator plate.

3. **NOTE:** *Inspect the crankshaft sensor ring for damage. If the crankshaft sensor ring has been dropped or has any visual damage, it must be discarded.*

Remove the crankshaft sensor ring.

4. Remove the oil pan. For additional information, refer to Engine Lubrication Components — Exploded View and Oil Pan in this section.
5. Disconnect the Crankshaft Position (CKP) sensor electrical connector.
6. Remove the bolt and the CKP sensor.
7. Using the Slide Hammer and the Crankshaft Rear Oil Seal Remover, remove and discard the crankshaft rear seal.



8. Remove the 6 bolts and the crankshaft rear seal retainer plate.

Installation

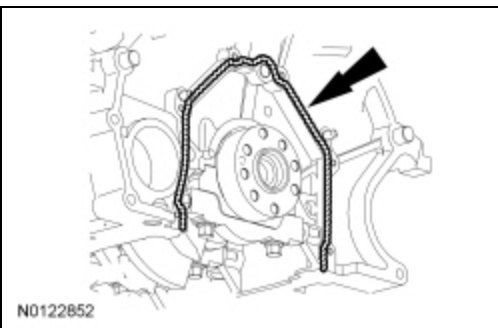
1. **NOTICE:** **Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

NOTE: *Clean the sealing surfaces with silicone gasket remover and metal surface prep. Follow the directions on the packaging.*

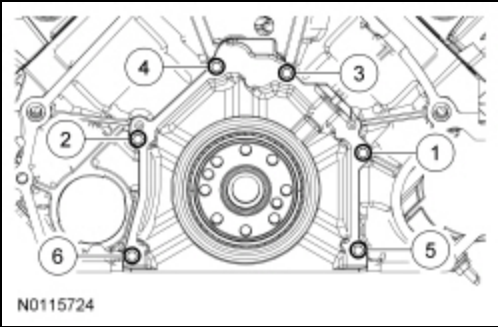
Clean and inspect the mating surface.

2. **NOTE:** *If the rear crankshaft seal retaining plate is not secured within 4 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure may cause future oil leaks.*

Apply a 3.75 mm (0.147 in) bead of silicone sealant to the rear crankshaft seal retainer mating surface on the engine block.

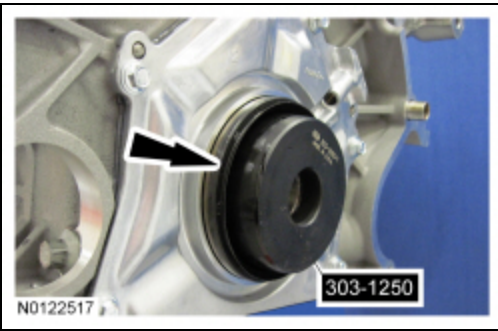


3. Install the crankshaft rear seal retainer plate and the 6 bolts, tighten in the sequence shown in 2 stages.
- Stage 1: Tighten to 10 Nm (89 lb-in).
 - Stage 2: Tighten an additional 45 degrees.

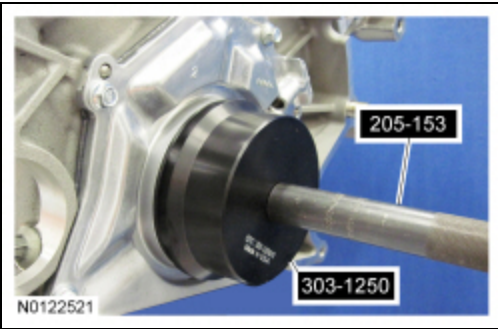


4. **NOTE:** *Lubricate the seal lips and bore with clean engine oil prior to installation.*

Position the Rear Main Seal Installer onto the end of the crankshaft and slide a new crankshaft rear seal onto the tool.



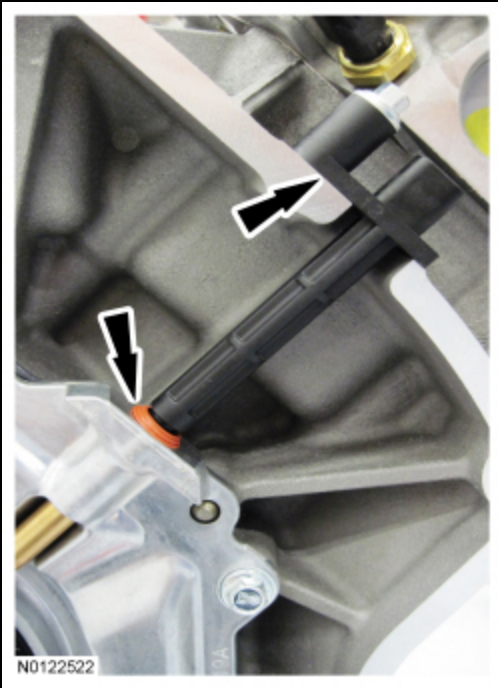
5. Using the Rear Main Seal Installer and Handle, install the new crankshaft rear seal.



6. **NOTICE:** The Crankshaft Position (CKP) sensor must be positioned into the fitting on the crankshaft rear seal retainer plate and be flush against the boss on the engine block before the bolt is installed. If the CKP sensor is installed incorrectly, the CKP sensor can be damaged.

Position the CKP sensor and install the bolt.

- Tighten to 10 Nm (89 lb-in).



7. Connect the CKP sensor electrical connector.
8. Install the oil pan. For additional information, refer to Engine Lubrication Components — Exploded View and Oil Pan in this section.
9. Install the crankshaft sensor ring.
10. Install the engine separator plate.
11. Install the flexplate. For additional information, refer to Flexplate or Flywheel and Crankshaft Rear Seal — Exploded View and Flexplate.